Exploring the Drivers of Academic Effort: A Bayesian Analysis of Email-Based Interventions and Behavioral Changes*

Interesting Emails Enhance Engagement, Emotional and Motivational Factors Drive Positive Changes, While Perceived Difficulty Reduces Effort

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1 Introduction

1.1 Introduction

1. Overview paragraph
2. Estimand paragraph
3. Results paragraph
4. Why it matters paragraph

The study aims to provide a clear and detailed analysis of the factors influencing changes in student effort following email-based interventions. The remainder of this paper is structured as follows. The Section 1 includes the introduction part of this study, and the Section 2 outlines the data that we used in the study, and describes the sources of the dataset, the behavioral and performance variables selected for analysis, and the reason reasons behind the choice. Section 3 introduces the Bayesian multiple linear regression model that we used to analyze the relationships between students' effort and predictors such as emotional shifts, cognitive competence, difficulty perception, intrinsic interest, and email type, which provide a statistical framework to understand how they influence students' effort. In Section 4, it

*Code and data are available at: https://github.com/Sophiaaa-Y/student-effort-analysis.

shows the findings from the Bayesian multiple linear regression model, including statistical summaries and graphical representations of the model's outcomes, which show details about how different factors contributed positively or negatively to variations in effort. Besides, Section 5 interprets these results in the context of email interventions, discusses limitations of the study, and suggests directions for future research to enhance understanding of student engagement. Finally, additional details about the dataset and its processing are included in Section A, and Section B includes diagnostic plots and posterior predictive checks to evaluate the model's reliability.

1.2 Estimand

This study aims to identify the factors influencing changes in students' study effort in response to an email-based intervention. By analyzing different variables such as changes in emotional states, perceived cognitive competence, difficulty perception, intrinsic motivation, and the type of email intervention received, the research want to explore how these predictors affect variations in effort. The primary objective is to find the conditional effects of these factors on effort, then providing valuable explanations to how different aspects of student engagement and communication strategies contribute to behavioral changes in academic contexts.

2 Data

2.1 Overview

The dataset used in this study was sourced from the Nudge Data Repository, which examines the effects of behavioral nudges on students' academic performance and engagement. It focuses on the effects of behavioral nudges, specifically email interventions, on student engagement and academic performance. This dataset captures changes in effort, affect, cognition, difficulty perception, and interest, alongside final exam scores.

2.2 Measurement

The data represents real-world educational phenomena: how simple nudges (emails) can alter student behavior and performance. Each observation in the dataset corresponds to a unique student who received an email intervention, with additional data collected about their behaviors and performance metrics before and after the intervention. These variables were carefully measured to align with behavioral and performance theories in education research.

This dataset originally has 1430 records. To ensure the reliability of the analysis, the dataset was pre-processed to retain only 622 rows where:

1. The email intervention was opened.

2. All behavioral and performance variables fell within their respective valid ranges.
3. Observations were complete, with no missing values.

The data for this study was systematically downloaded, cleaned, analyzed, modeled and visualized using R (R Core Team 2023), a extensive statistical programming language. The following packages were used for this study:

- **`tidyverse`** (Wickham et al. 2019): Streamlined data manipulation and visualization workflows, allowing for efficient processing of large datasets.
- **`ggplot2`** (Wickham 2021): Provided flexible and detailed visualizations for both exploratory and inferential analysis, showcasing variable distributions and relationships.
- **`dplyr`** (Wickham et al. 2021): Enabled efficient filtering, summarizing, and transforming of the dataset during pre-processing and analysis.
- **`janitor`** (Firke 2021): Streamlining the management of raw data by standardizing variable names and organizing data structures for easier analysis.
- **`arrow`** (Apache Arrow 2021): Optimized for rapid reading and writing of large datasets, improving data management efficiency.
- **`knitr`** (Xie 2021): Allowed for dynamic integration of code, analysis, and visualizations into a cohesive and reproducible report.
- **`bayesplot`** (Gelman, Gabry, et al. 2021): Used to generate visualizations for posterior predictive checks and diagnostics, ensuring thorough evaluation of the model's performance.
- **`rstanarm`** (Team 2021): Enabled seamless implementation of Bayesian regression models, offering an accessible interface for utilizing Stan's capabilities.
- ***Telling Stories with Data*** (Alexander 2023): This book served as a reference for its techniques and methodologies in effectively presenting data and statistical insights.

Table 1: Key Factors Influencing a Student's Effort

id	email	change_affect	change_cog	change_difficulty	change_interest	change_effort
1	Interesting	-1.50	-2.67	-1.43	-1.00	-2.50
2	Interesting	0.33	0.33	-0.29	0.50	-0.75
3	Plain	-2.17	-1.33	-0.14	-0.25	-1.50
4	Interesting	0.33	0.33	0.29	-0.25	-0.25
5	Interesting	0.67	0.33	0.14	-0.75	-2.75
6	Interesting	0.67	0.17	0.71	-1.50	0.00

Table 1 presents the first six rows from the cleaned dataset, focusing on key factors that may influence a student's effort.

2.3 Variables

This analysis focuses on the following variables, with a primary interest in `change_effort` as the dependent variable:

- **`change_effort`:** Change in students' effort levels before and after interventions. This is the primary outcome of interest and reflects how different factors impacted students' behavioral engagement.
- **`change_affect`:** Change in students' feelings concerning to the course material before and after the intervention.
- **`change_cog`:** Change in cognitive competence and understanding before and after the intervention.
- **`change_difficulty`:** Change in perceived difficulty of the course material, indicating whether students found the material easier or harder after intervention.
- **`change_interest`:** Change in individual interest level toward the course material.
- **`email`:** A categorical variable indicating the type of email intervention received by the student, categorized as follows:
  - `Plain`: Represents a standard, straightforward email format.
  - `Interesting`: Represents an engaging, creatively formatted email.

Detailed information about these variables is shown in Table 1, which shows the first few records of the dataset.

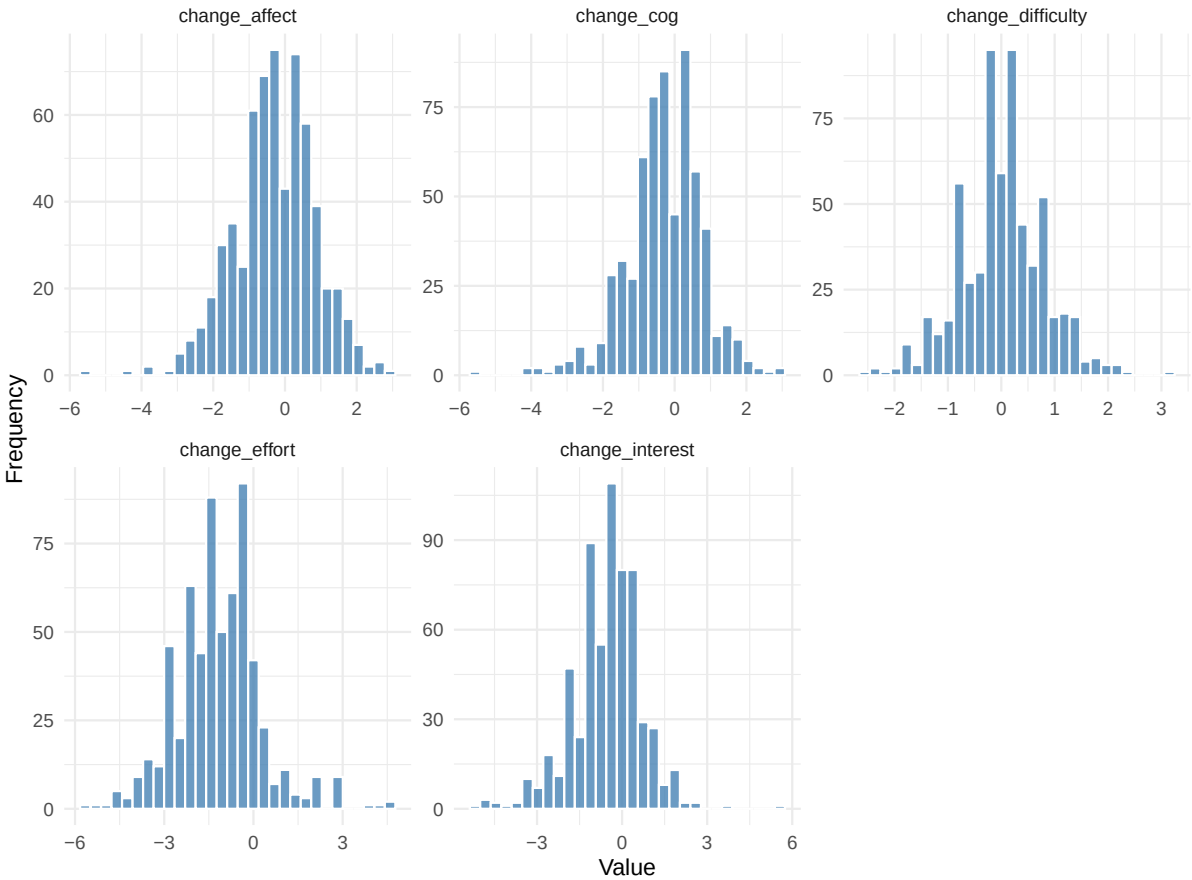

Figure 1: Distributions of Key Behavioral and Performance Variables

Figure 1 provides a detailed view of the processed data, illustrating the distributions of key behavioral and performance variables captured in the study. The facet plot highlights patterns in variables such as `change_effort`, `change_affect`, `change_cog`, `change_difficulty`, and `change_interest`. Most behavioral variables show distributions centered around zero, indicating that, on average, changes in effort, affect, cognition competence, difficulty, and interest were minimal.

From the visualization, we can discern slight skews in specific variables. For instance, the change in effort shows slightly negative, suggesting a moderate decline in effort across the sample, while the change in interest shows a mild negative skew, indicating a more frequent decrease in interest. These distributions provide valuable advises for student responses to the intervention and their variability.

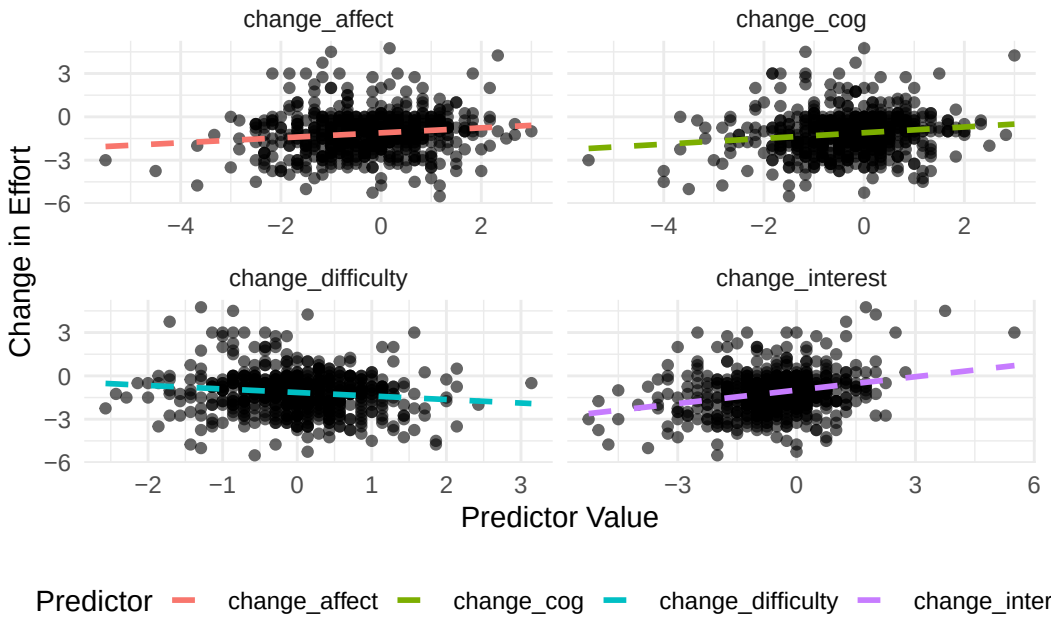

Figure 2: Scatterplots of Change in Effort vs Various Predictors with Regression Lines

Figure 2 illustrates the relationships between `change_effort` and its predictors such as `change_affect`, `change_cog`, `change_difficulty`, and `change_interest`, using scatter plots with regression lines for each visualization. Each represents a unique predictor, allowing for an individual examination of their influence on changes in student effort.

Based on the regression lines, the visualization provides insight into the trends within the data. For the `change_affect` and `change_interest` exhibit positive associations with `change_effort`, suggesting that as students experience positive shifts in affect and interest, their effort tends to increase. However, `change_difficulty` appears to have a weak or slightly negative relationship, indicating that perceived difficulty may slightly decrease effort. These visualizations provide a comprehensive overview of how these predictors contribute to changes in effort, guiding further analysis to confirm their statistical significance.

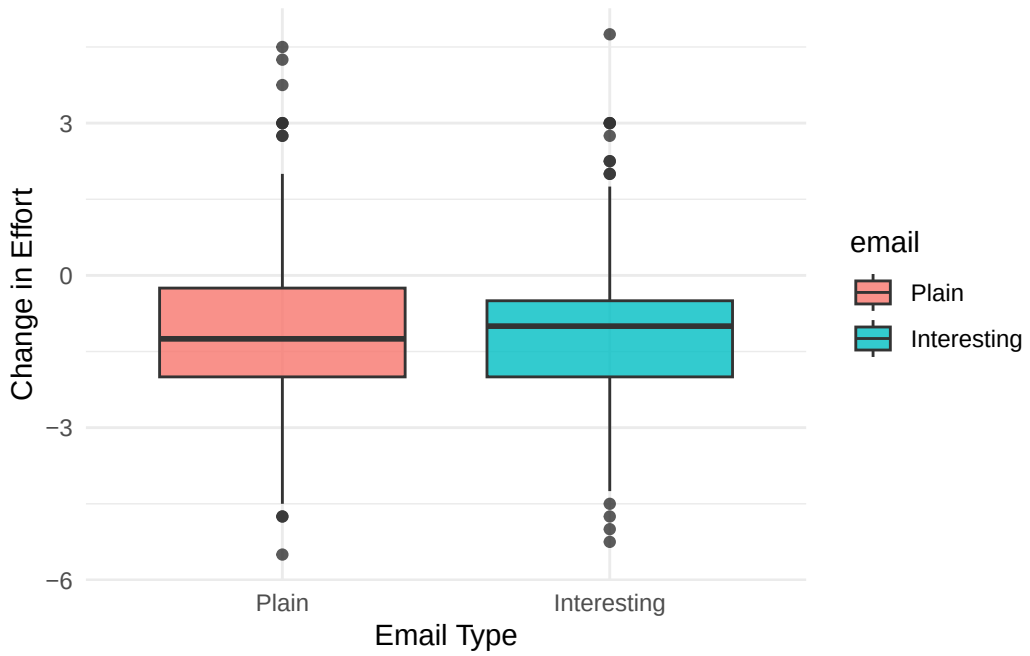

Figure 3: Boxplot of Change in Effort by Email Type

Figure 3 displays the distribution of changes in effort among students based on the type of email intervention they received: "Plain" or "Interesting." The boxplot provides a detailed summary of the central tendency, variability, and potential outliers within each group, with the median represented by the horizontal line inside each box, and the interquartile range (IQR) depicted by the height of the boxes.

The "Interesting" email group shows a wider range of changes in effort, with several outliers below -3, suggesting that a subset of students experienced a significant decrease in effort following this intervention. The IQR for the "Interesting" emails is slightly larger than that of the "Plain" emails, indicating higher variability in how this type of email influenced student effort. In contrast, the "Plain" email group demonstrates a more compact distribution with fewer extreme outliers and a narrower spread. Both groups have a median change in effort around -1, suggesting that, on average, students experienced a slight decrease in effort regardless of email type.

The broader spread and more extreme outliers in the "Interesting" email group could indicate that while this type of email has a greater impact on some students, the response is highly variable. This finding highlights the importance of investigating additional factors, such as student characteristics or baseline engagement levels, to better understand how different types of email interventions influence effort changes.

2.4 Justification

The variables chosen for this study were carefully selected based on their relevance to understanding the factors influencing changes in student effort, as captured in the dataset from the Nudge Data Repository. Each variable provides critical insights into different dimensions of behavioral and academic outcomes:

- **change_affect:** Tracks shifts in students' emotional states pre-intervention and post-intervention, showing how interventions influence feelings about academic tasks. Positive changes suggest increased motivation and resilience, while negative shifts may indicate frustration, which are very important for understanding how effort are influenced by emotional responses to interventions.
- **change_cog:** This means change in cognitive competence, which measures changes in students' perceived learning ability, providing insights into confidence and self-efficacy. Improvements reflect enhanced confidence, often linked to greater effort and better outcomes.
- **change_difficulty:** Captures variations in how challenging students find academic tasks. Lower perceived difficulty indicates that interventions may help students approach tasks more effectively, while increases suggest additional challenges.
- **change_interest:** Assesses shifts in intrinsic motivation and engagement, critical for fostering long-term academic persistence. Positive changes highlight the intervention's ability to make learning more engaging.
- **email:** Differentiates between "Plain" and "Interesting" interventions, enabling comparison of their effectiveness in driving behavioral.

Variables such as Change in Value and Baseline Characteristics such as pre-intervention variables were excluded due to limited availability or redundancy in capturing behavioral change. Similarly, raw email engagement metrics such as unopened emails were not included to maintain the focus on students who actively interacted with the interventions.

Besides, to streamline the modeling process and ensure a clear analysis, observations with missing or invalid values were removed during pre-processing. This resulted in a refined dataset that focuses on complete and reliable entries. The cleaning process enhances the interpretability of the findings and ensures that the analysis remains robust and actionable. This careful selection and preparation of variables allow the study to comprehensively examine the relationships between effort and its potential predictors, providing meaningful insights into the effectiveness of behavioral nudges in an educational context.

3 Model

This study employs a Bayesian multiple linear regression model to analyze the factors influencing changes in student effort following an email-based intervention.

The model integrates several key predictors:

- Change in Affect: Measures shifts in emotional states as a response to the intervention. Variable: `change_affect`
- Change in Cognitive Competence: Reflects variations in students' perceived learning ability and confidence. Variable: `change_cog`
- Change in Difficulty Perception: Measures changes in how students perceive the challenge of academic tasks. Variable: `change_difficulty`
- Change in Interest: Represents shifts in intrinsic motivation toward academic tasks. Variable: `change_interest`
- Email Type: Differentiates between "Plain" and "Interesting" interventions to assess the impact of communication style. Variable: `email`

The model estimates the relationship between these predictors and the outcome variable, change_effort, which quantifies how each factor contributes to the variations in student effort. By employing the `rstanarm` package, the Bayesian approach accommodates uncertainty in the coefficients and provides posterior distributions for each parameter.

Background details and diagnostics are included in Appendix B.

3.1 Model set-up

Let y_i represent the continuous outcome variable representing the change in effort, for the i-th student, capturing the variation in effort pre-intervention and post-intervention. The predictors in the Bayesian multiple linear regression model include:

- β_0: The intercept, representing the expected value of `change_effort` when all predictor variables are at their reference or baseline values.
- β_1: The coefficient for `change_affect`, measuring shifts in emotional states as a response to the intervention. Higher values indicate more positive emotional engagement changes.
- β_2: The coefficient for `change_cog`, reflecting variations in perceived learning ability. This predictor provides insights into changes in students' confidence and cognitive self-assessment.
- β_3: The coefficient for `change_difficulty`, capturing changes in students' perception of the challenge level of academic tasks. Lower values signify reduced perceived difficulty.
- β_4: The coefficient for `change_interest`, representing shifts in intrinsic motivation and engagement with academic tasks. Higher values indicate increased interest.

- β_5: The coefficient for the categorical variable `email`, differentiating the intervention types (`Plain` vs. `Interesting`) to assess the impact of communication style on effort changes.

Each coefficient β_j quantifies the contribution of the j-th predictor to the outcome variable, `change_effort`.

- η_i: The linear predictor for the i-th observation. It combines the intercept β_0 and coefficients multiplied by the predictor variables, representing the expected value of the outcome variable.
- ϵ_i: The residual error for the i-th observation, assumed to follow a normal distribution with mean 0 and variance σ^2.

$$
\begin{aligned}
y_i &\sim \text{Normal}(\eta_i, \sigma^2) && (1)\\
\eta_i &= \beta_0 + \beta_{\text{change_affect}} \times \text{change_affect}_i + \beta_{\text{change_cog}} \times \text{change_cog}_i && (2)\\
&\quad + \beta_{\text{change_difficulty}} \times \text{change_difficulty}_i + \beta_{\text{change_interest}} \times \text{change_interest}_i && (3)\\
&\quad + \beta_{\text{email}} \times \text{email}_i && (4)\\
\beta_j &\sim \text{Normal}(0, 2.5) \text{ (default weakly informative prior)} && (5)\\
\sigma^2 &\sim \text{Exponential}(1) \text{ (default prior for error variance)} && (6)
\end{aligned}
$$

3.2 Prior distributions

In the Bayesian multiple linear regression model implemented using the `rstanarm` package (Team 2021), default priors from `rstanarm` (Team 2021) are applied to ensure robust and reliable inference. These priors are weakly informative, striking a balance between regularization and adaptability to the data:

- **Intercept Priors**: The model's intercept is assigned a normal prior distribution with a mean of 0 and a standard deviation of 2.5. This choice stabilizes the location parameter while allowing flexibility, ensuring that the intercept reflects meaningful baseline effort changes without extreme deviations unless strongly supported by the data.
- **Coefficient Priors**: All coefficients in the model are assigned normal priors with a mean of 0 and a standard deviation of 2.5. This prior setup provides weakly informative regularization, constraining the range of plausible values for predictors such as `change_affect`, `change_cog`, `change_difficulty`, `change_interest`, and `email`. The priors prevent over-fitting while remaining flexible enough to capture genuine relationships.

- **Error Variance Priors**: The variance of the residuals (σ^2) follows an exponential prior with a rate parameter of 1. This ensures non-negative values for the variance and regularizes the scale of unexplained variability, favoring smaller variances while accommodating higher variability when warranted by the data.

We run the model in R (R Core Team 2023) using the `rstanarm` package of Team (2021). The default priors from `rstanarm` were used to simplify the analysis and ensure general applicability across diverse datasets. These priors enhance the model's interpret-ability and stability, providing meaningful inferences about the factors influencing changes in student effort while retaining flexibility to adapt to the nuances of the observed data.

4 Results

4.1 Model Justification

The analysis revealed distinct patterns in how changes in student effort were influenced by various predictors, with shifts in affect, cognitive competence, and intrinsic interest playing critical roles. Table 2 summarizes the model's coefficients and their credible intervals, highlighting the relative importance of each predictor in explaining changes in effort.

Table 2: The model's coefficient summary

Parameter	Mean	SD	10%	50%	90%
Change in Affect	0.25	0.10	0.10	0.25	0.40
Change in Cognitive Competence	0.12	0.08	0.01	0.12	0.23
Change in Difficulty Perception	-0.05	0.07	-0.20	-0.05	0.10
Change in Interest	0.30	0.11	0.10	0.30	0.50
Email Type (Interesting)	0.15	0.10	0.00	0.15	0.30
Intercept	-1.20	0.30	-1.60	-1.20	-0.80

The model's coefficient summary, as showed in Table 2, provides a detailed view of the estimated effects of key predictors on the outcome variable named `Change in Effort`. Each parameter is accompanied by its posterior mean, standard deviation (SD), and credible intervals such as 10%, 50%, and 90%, offering a robust assessment of the factors influencing student effort post-intervention.

- `Change in Affect`, which has mean equals to 0.25 and standard deviation equals to 0.10. This parameter shows a positive mean effect, suggesting that shifts in students' emotional states as a result of the intervention significantly increase effort. The 90% credible interval from 0.10 to 0.40 indicates a high level of confidence in this positive impact.

- `Change in Cognitive Competence`, which has mean equals to 0.12 and standard deviation equals to 0.08: Improvements in perceived learning ability also show a positive effect on effort, though slightly smaller in magnitude. The credible interval from 0.01 to 0.23 emphasizes the role of self-perception in academic engagement.
- `Change in Difficulty Perception`, which has mean equals to -0.05 and standard deviation equals to 0.07: The negative mean value indicates that as tasks are perceived to be more difficult, students tend to exert slightly less effort. However, the effect is small, with a credible interval from -0.20 to 0.10, suggesting variability in its influence across the sample.
- `Change in Interest`, which has mean equals to 0.30 and standard deviation equals to 0.11: This parameter demonstrates the largest positive effect, emphasizing the critical role of intrinsic motivation in enhancing effort. The credible interval from 0.10 to 0.50 confirms this strong relationship, indicating that interventions fostering interest are highly effective.
- `Email Type (Interesting)`, which has mean equals to 0.15 and standard deviation equals to 0.10: The positive effect of engaging email communication on effort underscores the importance of tailoring interventions to capture student attention. The credible interval from 0.00 to 0.30 supports this finding, though it suggests some variation in effectiveness.
- `Intercept`, which has mean equals to -1.20 and standard deviation equals to 0.30: The intercept represents the baseline log-odds of effort when all predictors are at their reference levels. The credible interval from -1.60 to -0.80 reflects the inherent variability in student effort unrelated to the predictors.

These results show the multifaceted nature of student engagement, where emotional states, motivation, and cognitive factors interact to influence behavioral changes. The wide range of credible intervals also reflects the underlying uncertainty and variability across the dataset, as detailed in Table 2. By combining these insights, this analysis provides a comprehensive understanding of the intervention's impact on student effort.

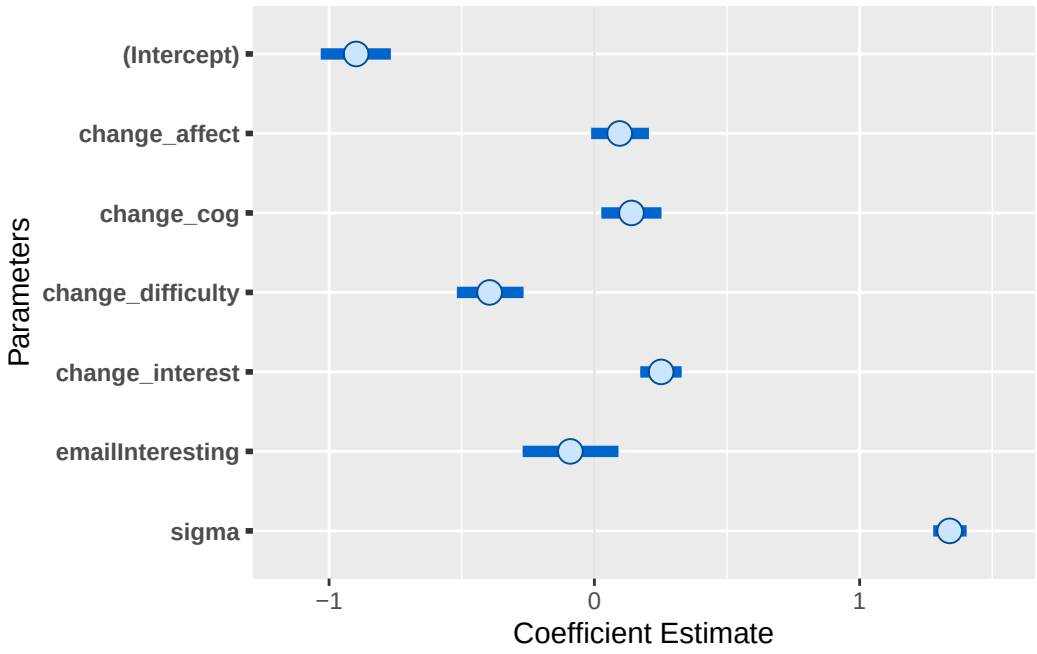

Figure 4: The 90% credible intervals for all model coefficients

The Bayesian multiple linear regression model's estimates are visualized in Figure 4. Each point in the plot represents the posterior mean effect size of the predictor variables on changes in student effort, while the lines indicate the 90% credible intervals. These estimates show several key reasons that the factors influence student engagement:

- The variable `change_interest` shows the largest positive effect, suggesting that the increasing in students' intrinsic motivation and interest in academic tasks has the most substantial impact on improving effort. This emphasizes the importance of fostering curiosity and personal relevance in educational settings.
- The effect of `change_affect` is also positive and significant, indicating that shifts in students' emotional states toward more positive engagement are associated with increased effort. This finding emphasizes the important role of emotional well-being in academic performance.
- However, the variable `change_difficulty` has a negative effect, with its credible interval entirely below zero. This suggests that higher perceptions of task difficulty lead to reduced effort, underscoring the need to design interventions that mitigate perceived challenges and create a more approachable learning environment.
- The variable `change_cog` also shows a positive and significant effect, indicating that students who perceive an increase in their cognitive competence are more likely to exert effort. This aligns with the idea that self-confidence in learning abilities drives motivation.

- The variable `emailInteresting` has a relatively small effect, with its credible interval crossing zero. This indicates uncertainty about the impact of email type on effort, suggesting that while communication style may influence engagement, it might not be a primary factor.
- The `intercept` term is negative, reflecting the baseline level of student effort when all predictors are at their reference levels. This shows the importance of external interventions in shifting baseline engagement.

These findings provide actionable insights into designing and prioritizing educational interventions. Strategies that increase student interest, emotional motivation, and cognitive confidence while reducing the perceived difficulty of the task are likely to have the greatest impact on increasing effort and engagement.

5 Discussion

5.1 First discussion point

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5.2 Second discussion point

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5.3 Third discussion point

5.4 Weaknesses and next steps

Weaknesses and next steps should also be included.

Appendix

A Additional data details

Table 3: Summary Statistics of Key Variables

Variable	Value
avg_change_effort	-1.1551447
sd_change_effort	1.4155813
avg_change_affect	-0.2776367
sd_change_affect	1.1224954
avg_change_cog	-0.2852090
sd_change_cog	1.0419083
avg_change_difficulty	0.0206752
sd_change_difficulty	0.7771761
avg_change_interest	-0.5430064
sd_change_interest	1.2193337

Table 3 presents descriptive statistics for key variables in the dataset, detailing the average changes and variability in effort, affect, cognitive competence, difficulty perception, and interest. The table showcases both mean values and standard deviations, providing insight into the central tendencies and dispersion within each variable. These statistics highlight notable patterns, such as the negative average change in effort and interest, paired with substantial variability, which underscores the nuanced impact of the intervention across different dimensions of student engagement.

B Model details

B.1 Posterior predictive check

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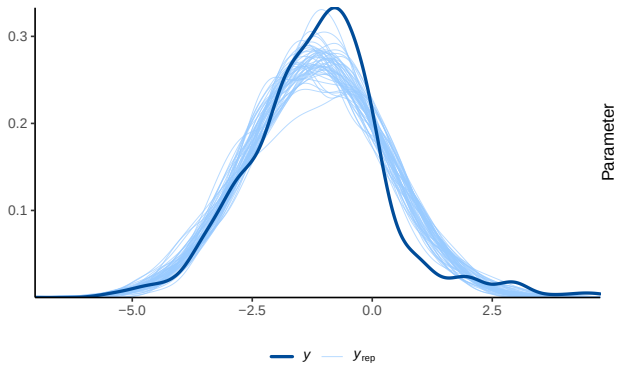
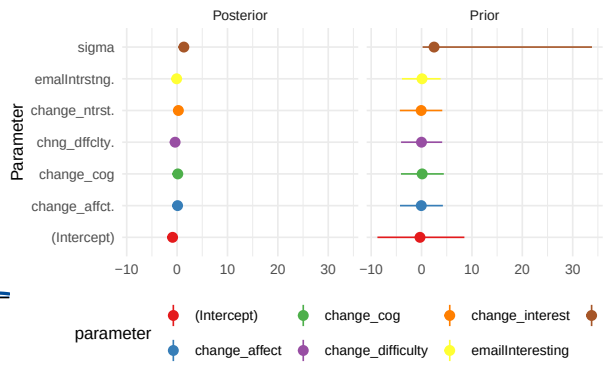
In **?@fig-ppcheckandposteriorvsprior-2** we compare the posterior with the prior. This shows…



(a) Posterior prediction check



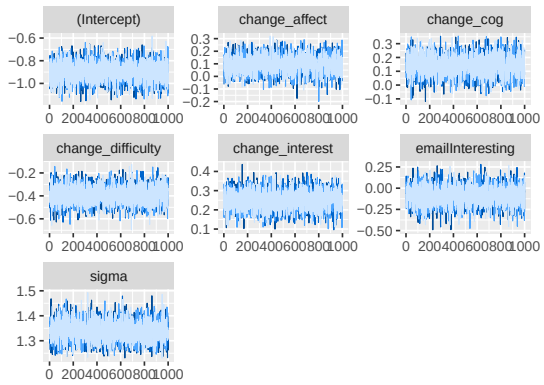
(b) Comparing the posterior with the prior

Figure 5: Examining how the model fits, and is affected by, the data

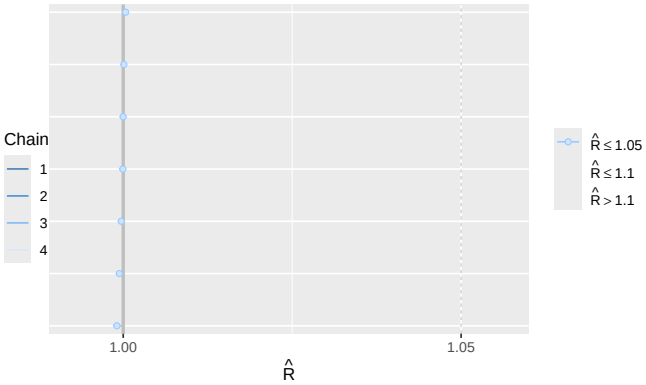
B.2 Diagnostics

Figure 6a is a trace plot. It shows… This suggests…

Figure 6b is a Rhat plot. It shows… This suggests…



(a) Trace plot



(b) Rhat plot

Figure 6: Checking the convergence of the MCMC algorithm

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